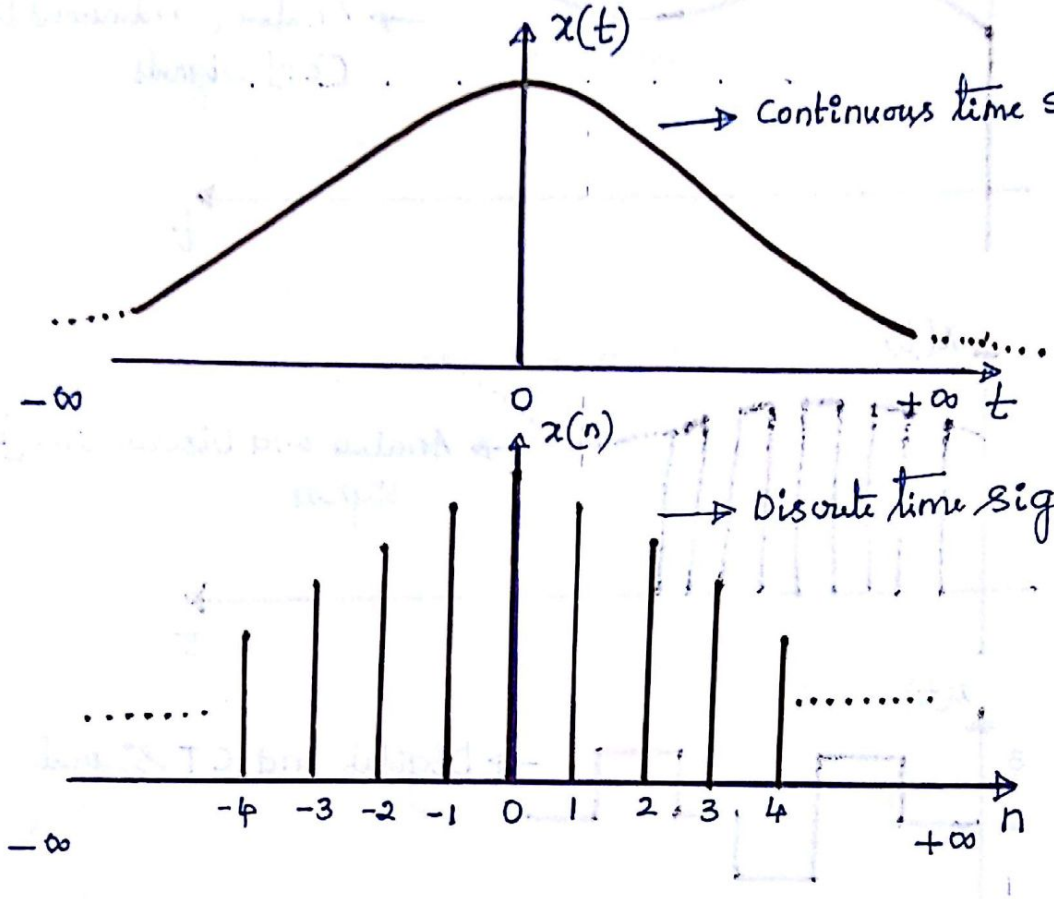
Dr. K.B. Ramesh ADDC

Digital Electronics and Microprocessors - (DEMP)

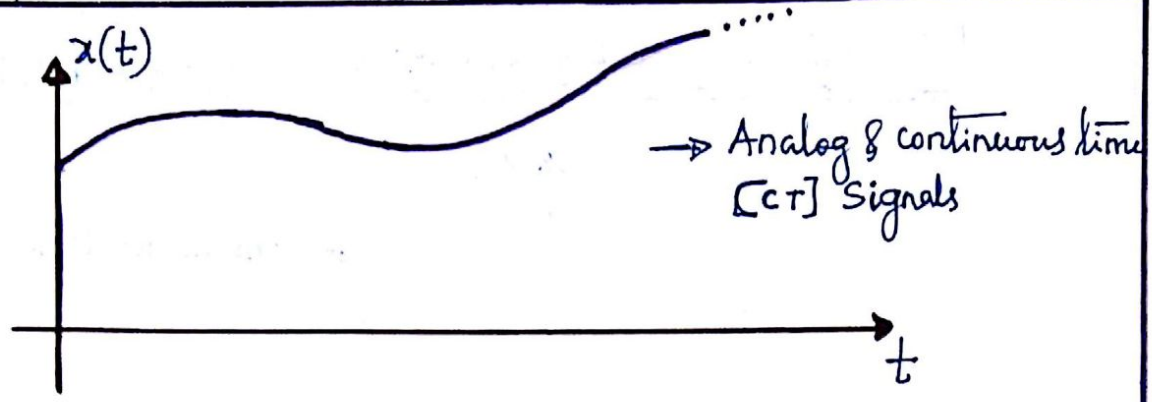


* Classification of Signals:

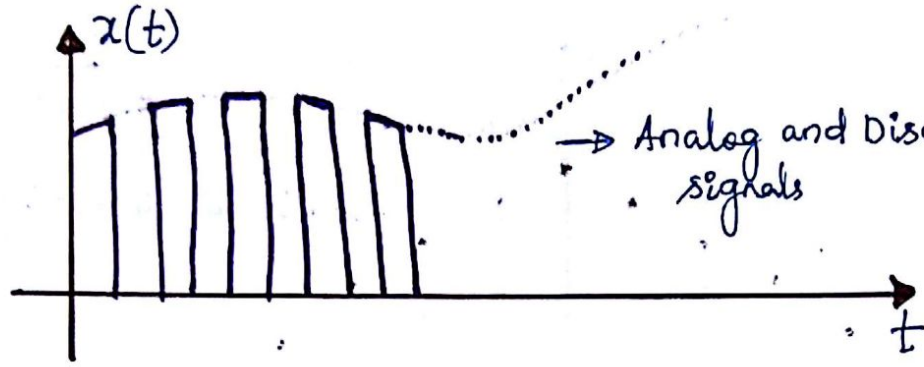
→ If Y-axis is infinite amplitude Values then it is analog signals
→ If Y-axis is finite amplitude Values then it is Digital signals
→ If X-axis is continuous time then it is continuous-time signals
→ If X-axis is discrete time then it is Discrete-time signals.

NOTE: Analog / Digital ⟶ Amplitude (Y-Axis)
continuous / Discrete ⟶ Time (X-Axis)

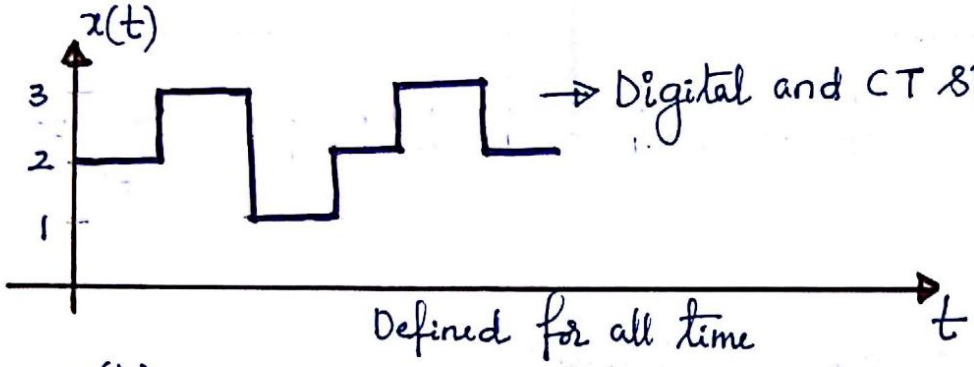
Scanned by CamScanner



→ Analog & continuous time [CT] Signals

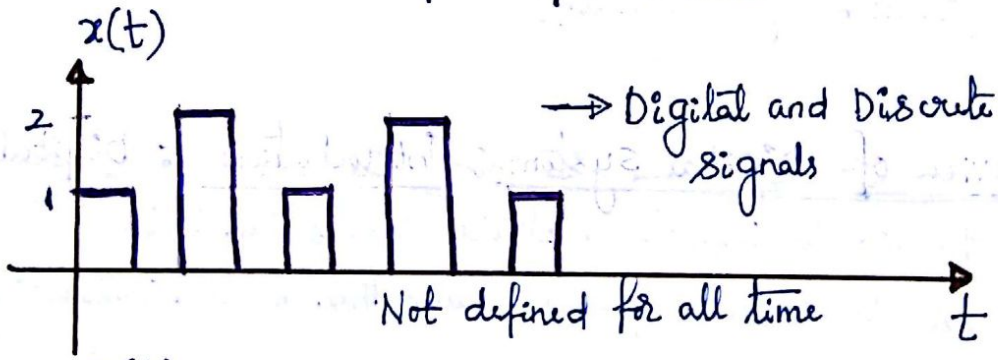


→ Analog and Discrete time [DT] signals



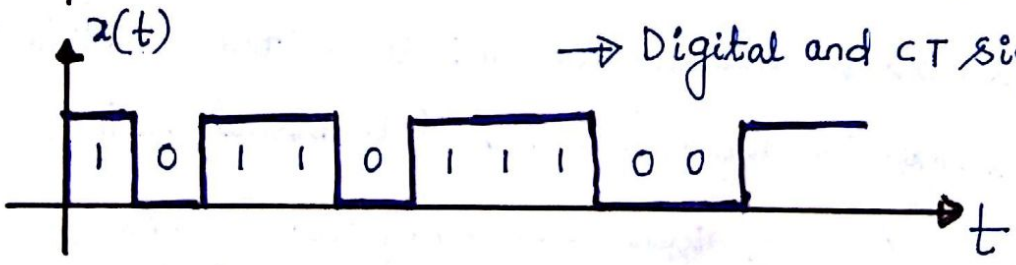
→ Digital and CT signal

Defined for all time



→ Digital and Discrete time [DT] signals

Not defined for all time



→ Digital and CT signal

NOTE: Discrete time signal is the subset of analog signal.

Scanned by CamScanner

Dr. KBR

* Digital Logic Gates:

→ A Digital Logic Gate is an electronic circuit which makes logical decisions based on the combination of digital signals present on its inputs.

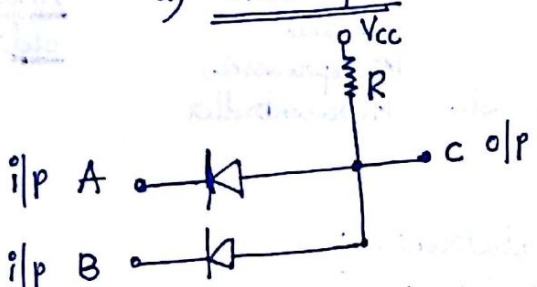
* How are logic gates constructed?

→ Logic gates can be constructed in many ways.

(1) The simplest is diode Logic [DL]:

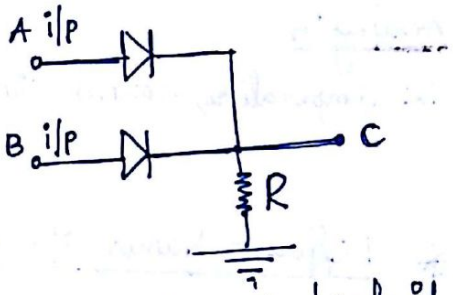
→ uses diodes and Resistors

a) AND Gate



output voltage is high only if both A and B are high.

b) OR Gate



output voltage is high if either (or both) A and B are high.

(2) Resistor - transistor Logic [RTL]:

→ uses Bipolar junction transistors [BJTs]

(3) Diode - transistor Logic [DTL]:

→ uses diodes in place of the resistors.

(4) Transistor - transistor logic [TTL]:

→ Basis for the popular 7400 TTL logic series.

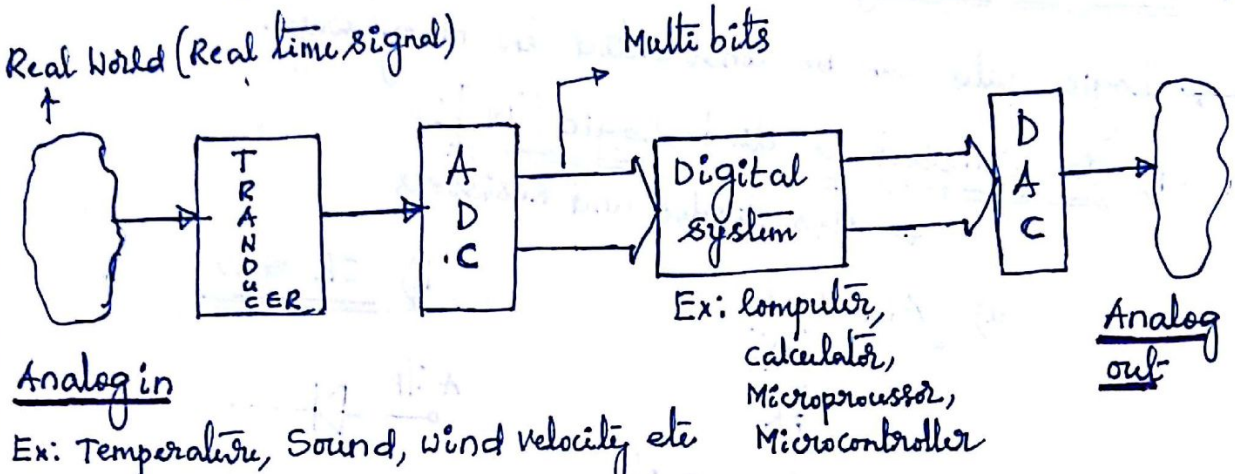
Scanned by CamScanner

* Finally we can conclude as follows:

1) Analog and Digital Signals

2) Continuous time (CT) and Discrete time (DT) Signals.

* Block diagram:



* Different Names of Digital Electronics:

→ Logic Design
→ Digital circuits
→ Digital logic

* A digital circuit is typically constructed from small electronic circuits called logic gates.

* Digital electronic circuits operate with voltages of two logic levels namely logic low and logic high.

* Logic gates → Basic gates: AND, OR & NOT.
→ universal gates: NAND & NOR
→ Special gates: Ex-OR & Ex-NOR

Scanned by CamScanner

Dr. K.B. Ramesh

* NOTE :-

> We live in an analog world
> Colors we see, heat we feel,
> Sounds we make and hear are all
> in the form of analog signals.

* Summary:

→ An analog signal can take any value within a given range of time.

→ A digital signal can only represent one of a discrete set of values.

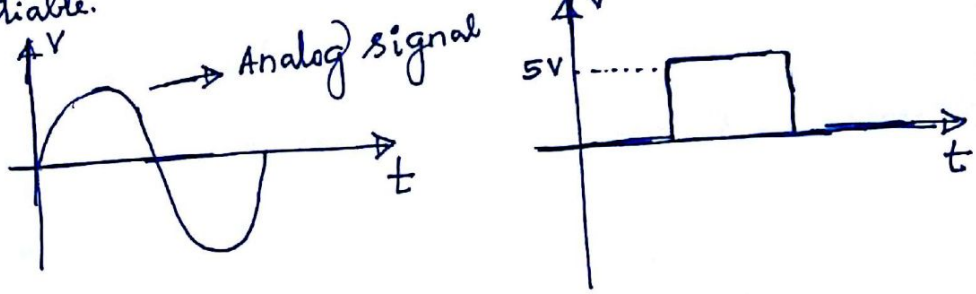
→ A digital signal which can have only two discrete values is called a digital signal. Ex: Square wave

→ Digital circuit handles only a digital signal.

→ Analog signal is a continuously varying signal.

→ Analog signal is less Reliable and Digital signal is more Reliable.

Ex:



Scanned by CamScanner

→ An Integrated circuit (IC) is an electronic device that gathers (or Integrates) a number of electronic components on a small semiconductor chip.

→ IC's → Analog IC: Handle Continuous signals [Linear IC]
→ Digital IC: Handle discrete signals such as binary values.

Scanned by CamScanner

5) CMOS Logic:

→ used in computers

→ uses almost no power except when it is switching and also it has high noise immunity.

* Standard commercial digital logic gates are available in two basic families or forms.

1) Transistor - Transistor Logic [7400 series]: TTL

2) complementary Metal - oxide - silicon [CMOS]; 4000 Series of chips.

* The Notation of TTL / CMOS refers to the logic Technology used to manufacture the Integrated circuit (IC) or a chip.

* TTL IC's use NPN and PNP type Bipolar Junction Transistors.

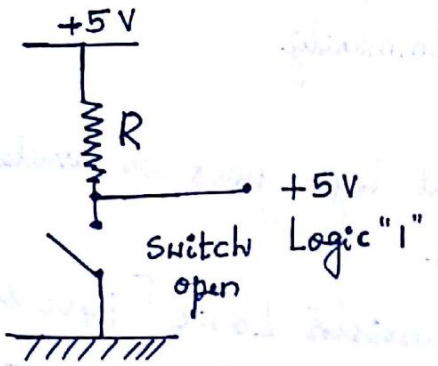
* CMOS Ic's use complementary MOSFET

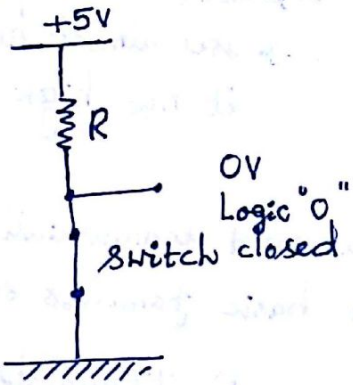
* classification of Integrated circuits:

1) Small scale Integration (SSI):
2) Medium scale Integration (MSI):
3) Large scale Integration (LSI):
4) Very - Large scale Integration (VLSI):
5) Super - Large scale Integration (SLSI):
6) Ultra - Large scale Integration (ULSI):

Scanned by CamScanner

* Ideal Digital Logic gate Voltage levels:





Scanned by CamScanner

ಶೆ.ವಿ.ರ ಪ್ರೀತಿ

* Subtraction by 2's complement:

Examples:

1) 10110

⊖) 11010 —1's→ 00101 —+1→

+1 00001

00110

10110

11100 → Again take 2's complement

00011

+ 1

00100

2) 110110

⊖) 010110 —1's→ 101001

+1 → 000001

101010

110110

1 100000

drop ← complement

64 32 16 8 4 2 1

46
14
32

3) $(46)_{10} - (14)_{10} = ?$

$(46)_{10} + [-(14)_{10}]$ ← 2's complement

$(46)_{10} = 101110$

$(14)_{10} = 001110$ —1's complement→

110001

+ 000001

100010

101110

Neglect 1 100000 $= (32)_{10}$

Scanned by CamScanner

KBR

* Binary Addition :

Rules : 0 + 0 = 0
0 + 1 = 1
1 + 0 = 1
1 + 1 = 0, 1 in carry

Example :
```
  0 1 0 1
+ 1 0 1 0
---------
  1 1 1 1
```
①

```
  1
  0 1 0 1
+ 0 1 1 0
---------
  1 0 1 1
```
②

8-bit No:

```
    1   1
  0011 1001
+ 0010 1001
-----------
0110 0010
```
③

```
  0 0 1 1
+ 0 0 1 1
---------
  0 1 1 0
```
④

```
  1 1 0
  0 1 1 0
+ 0 1 1 1
---------
1 1 0 1
```

00 → 0
01 → 1
10 → 2
11 → 3

0 + 1 = 01
1 + 1 = 2 = 10
1 + 1 + 1 = 3 = 11

* Binary Subtraction :

0 − 0 = 0
1 − 1 = 0
1 − 0 = 1
0 − 1 = 1, Borrow of 1

Example : (Borrow 1 means 2)
```
  2
  1 0 1
- 0 1 1 1
---------
  0 1 1 0
```

Example :
```
    2   2   2
  1 0 1 0 0 1 0
-   1 1 1 1 1 1
-----------------
  0 0 1 0 0 1 1
```

Example :
```
    1
  0 2 2
  1 0 0
-   1 1
---------
  0 0 1
```

Example :
```
    1   1   1   2
  2   2   2   2
  1 0 0 0 0 1
-   1 1 1 1
-------------
  0 1 0 0 1 0
```

Example :
```
    1       2
  0 2 2 0 2
  1 1 0 0 1 0 1 1
-   1 0 1 0 1 1 0
-----------------
  0 1 1 1 0 1 0 1
```

Scanned by CamScanner

Example :

① 1 0 0 0 — 8
(−) 0 1 1 — 3
0 0 0 1 — 5

② 1 0 0 0 — 8
(−) 0 0 1 1 — −3
0 1 0 1 — 5

③ 1 0 0 0 0 — 16
− 0 1 1 1 — −7
0 1 0 0 1 — 9

④ 1 0 1 0 0 → 20
0 0 1 1 0 → −6
0 1 1 1 0 — 14

⑤ 1 0 0 0 1 → 17
(−) 0 1 0 1 1 → 11
0 0 1 1 0 → 06

NOTE : Addition & Subtraction of binary Numbers.

Scanned by CamScanner

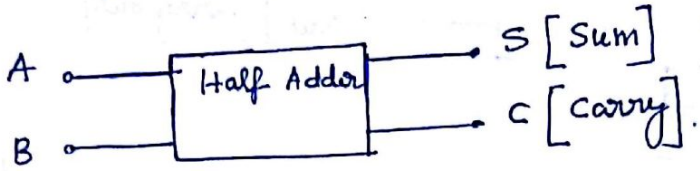
Dr. K.B. Ramesh

* Combinational logic circuits:

→ Designing of Half-Adder:

→ It adds two single bit of Numbers.

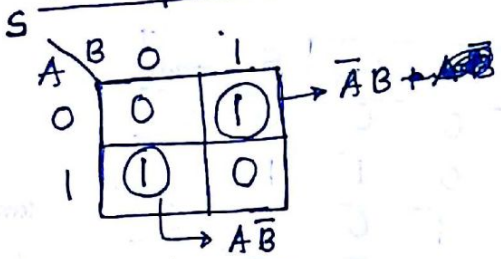
→ The half adder circuit has two inputs, which add two input digits and generate a carry and Sum.



Truth Table

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

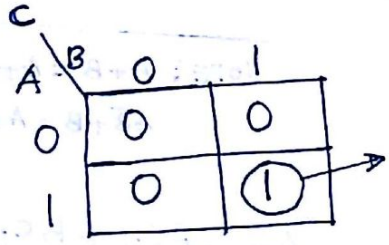
K-map for Sum



$Sum = A\bar{B} + \bar{A}B$

$= A \oplus B$ → Ex-OR operation

K-map for carry

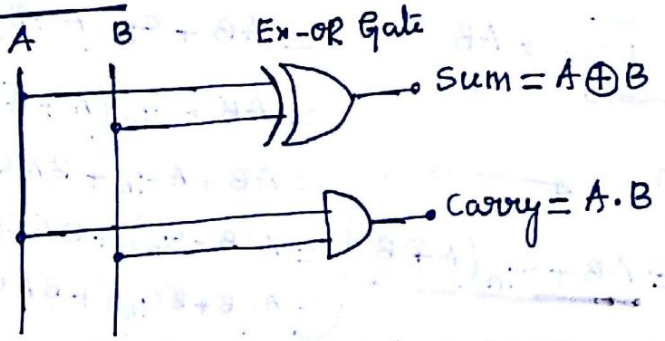


$carry = AB$

NOTE

A	B	$A\cdot\bar{B} + B\cdot\bar{A}$	S
0	0	0·1 + 0·1	0
0	1	0·0 + 1·1	1
1	0	1·1 + 0·0	1
1	1	1·0 + 1·0	0

Logic circuit



Summary: Ex-OR

if both the i/p's are same then o/p is zero

if both the i/p's are not same then o/p is one.

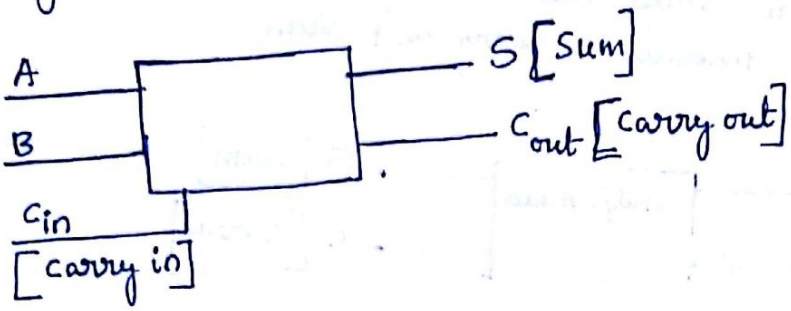
Scanned by CamScanner

ಡಿ.ವಿ.ಡಿ ಟೆಕ್

* Designing of Full Adder :

→ A full adder adds two inputs and a carried input from another adder and gives a two bit output i.e Sum and carry.

→ Designed to add two single digits with a carry.



NOTE:
(A) + (B) + (C_{in}) =
1 + 1 → 0 1
↓ ↓
S C

Truth Table

A	B	C_{in}	S.	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

K-map for Sum

Sum

A \ B C_{in}	00	01	11	10
0	0	1	0	1
1	1	0	1	0

Sum = $A \oplus B \oplus C_{in}$

NOTE: $A + B = A + \bar{A}B$

$A\bar{B} + B = A + B$

K-map for C_{out}:

C_{out}

A \ B C_{in}	00	01	11	10
0	0	0	1	0
1	0	1	1	1

BC_{in}, AB, AC_{in}

$= AB + A\bar{B}C_{in} + B\bar{A}C_{in}$

$= AB + C_{in}(A\bar{B} + B\bar{A}) = AB + C_{in}(A \oplus B)$

$C_{out} = AB + AC_{in} + BC_{in}$

$= AB + C_{in}(A + B)$

$= AB + C_{in}(A + \bar{A}B)$

$= AB + AC_{in} + B\bar{A}C_{in}$

$= A(B + C_{in}) + B\bar{A}C_{in}$

$= A(B + \bar{B}C_{in}) + B\bar{A}C_{in}$

Scanned by CamScanner

Dr. K. B. Ramesh

$$Sum = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in}$$

$$= C_{in}\left[AB + \bar{A}\bar{B}\right] + \bar{C}_{in}\left[A\bar{B} + B\bar{A}\right]$$

$$= C_{in}\left[\overline{A \oplus B}\right] + \bar{C}_{in}\left[A \oplus B\right]$$

(C_{in} = A, $\overline{A \oplus B}$ = $\bar{B}$, $\bar{C}_{in}$ = $\bar{A}$, $A \oplus B$ = B)

$$=$$

$$Sum = C_{in} \oplus A \oplus B$$

NOTE

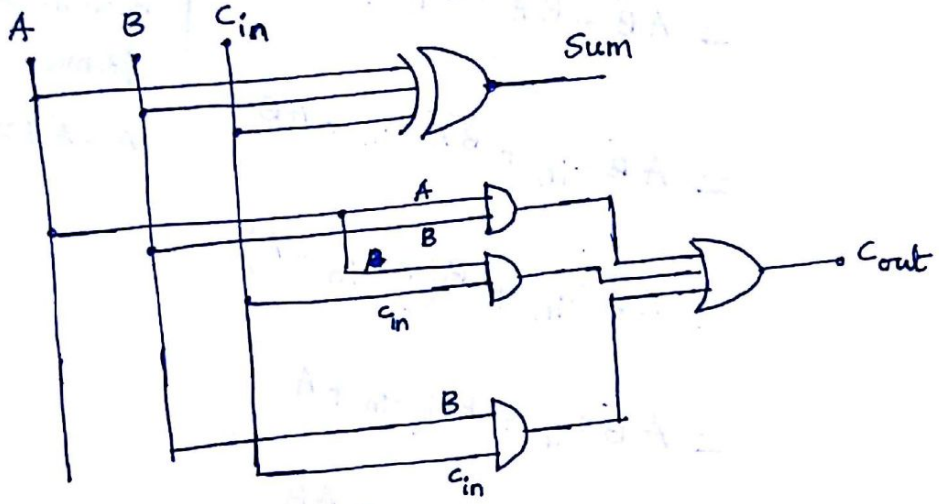
Ex-NOR gate



$y = \overline{A \oplus B}$

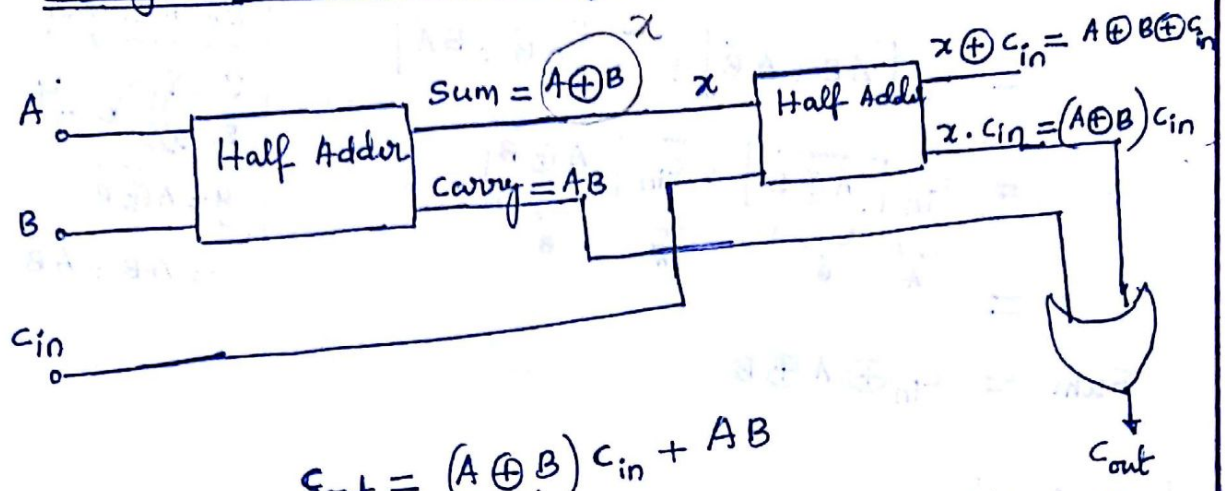
$= AB + \bar{A}\bar{B}$

Logic circuit



Scanned by CamScanner

* Design of Full adder using Half Adders :



$$C_{out} = (A \oplus B) C_{in} + AB$$

$$= (A\bar{B} + B\bar{A}) C_{in} + AB$$

$$= A\bar{B} C_{in} + B\bar{A} C_{in} + AB$$

$$= A\bar{B} C_{in} + B(\bar{A} C_{in} + A)$$

$$= A\bar{B} C_{in} + B(C_{in} + A)$$

$$= A\bar{B} C_{in} + BC_{in} + AB$$

$$= C_{in}(A\bar{B} + B) + AB$$

$$= C_{in}(A + B) + AB$$

Boolean Theorem formula

$A + \bar{A}B = A + B$

$$C_{out} = AB + AC_{in} + BC_{in}$$

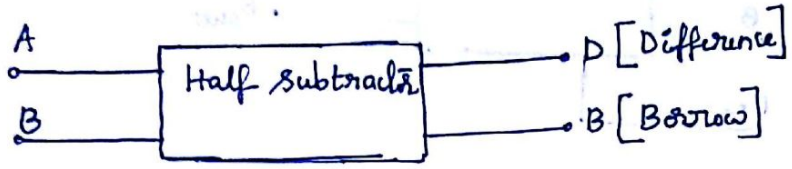
Scanned by CamScanner

Dr. KBR

* Half Subtractor :

→ is a combinational circuit which is used to perform subtraction of two bits.

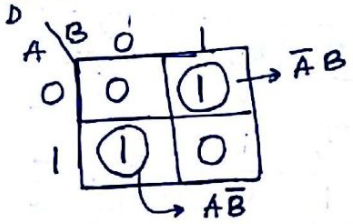
→ It has two inputs, the minuend and subtrahend and two outputs the difference and borrow out.



Truth Table :

A	B	D	B
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

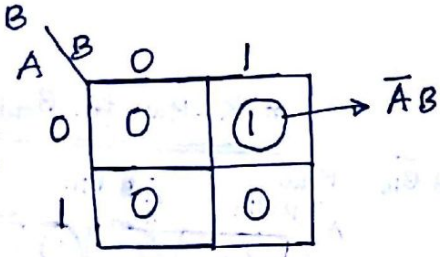
K-Map for Difference



$D = A\bar{B} + B\bar{A} = A \oplus B$

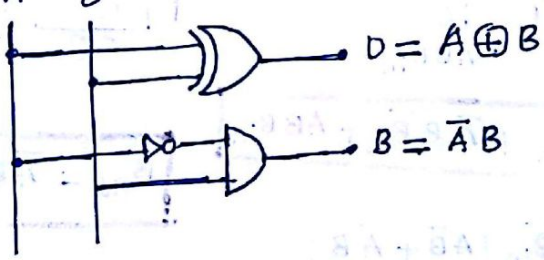
NOTE:
- Total diag
- not covering of the group leads to Ex-OR

K-Map for Borrow :



$Borrow = \bar{A}B$

Logic circuit :

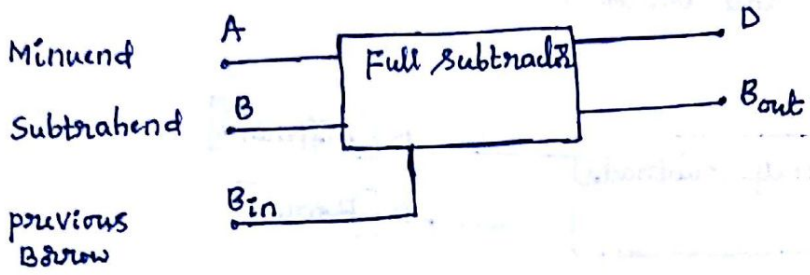


Scanned by CamScanner

* Full Subtractor :

→ performs subtraction of two bits, one is minuend and other is subtrahend, taking into account borrow of the previous minuend bit.

→ The circuit has three inputs and two outputs.



Drawback of Half Subtractor

$A \rightarrow 0\ 0\ 1$
$B \rightarrow 0\ 1\ 0$
$1\ 1\ \times$

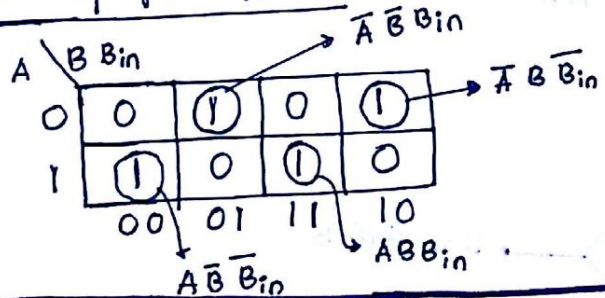
* Truth Table :

A	B	B_{in}	D	B_{out}
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

A − B − B_{in} : 0 1 11

0 − 0 = 0 ; 0 − 1 = 11 ; 0 − 1 = 11
1 − 0 = 1 ; 0 − 1 = 11 ; 1 − 1 = 10
1 − 0 = 1 ; 1 − 1 = 0
0 − 1 = 11

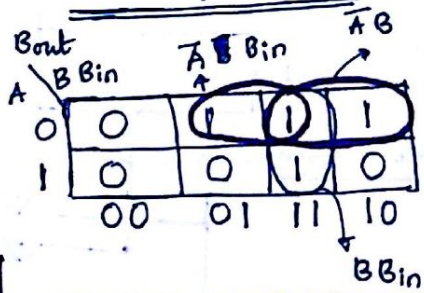
* K-Map for Difference :



$$D = A\bar{B}\bar{B}_{in} + \bar{A}\bar{B}B_{in} + \bar{A}B\bar{B}_{in} + ABB_{in}$$

$$D = B_{in}(AB + \bar{A}\bar{B}) + \bar{B}_{in}(A\bar{B} + \bar{A}B)$$

* K-Map for B_{out}



$$B_{out} = \bar{A}B_{in} + \bar{A}B + BB_{in}$$

Scanned by CamScanner

Dr. K. B. Ramesh

$$D = B_{in}\left(\overline{A \oplus B}\right) + \overline{B}_{in}\left(A \oplus B\right)$$

(B_{in}: A; $\overline{A \oplus B}$: $\overline{B}$; $\overline{B}_{in}$: $\overline{A}$; $A \oplus B$: B)

$$D = B_{in} \oplus (A \oplus B)$$

NOTE: $\oplus$ → Ex-OR operation

$$x = A \oplus B = A\overline{B} + B\overline{A}$$

$$\overline{x} = \overline{A \oplus B} = AB + \overline{A}\,\overline{B}$$

Formulae

$$\overline{x+y} = \overline{x} \cdot \overline{y}$$

$$\overline{xy} = \overline{x} + \overline{y}$$

We have,

$$A \oplus B = A\overline{B} + B\overline{A}$$

$$\overline{A \oplus B} = \overline{A\overline{B}} + \overline{B\overline{A}}$$

($A\overline{B}$: x; $B\overline{A}$: y)

$$= \overline{A\overline{B}} \cdot \overline{B\overline{A}}$$

$$= \left(\overline{A} + \overline{\overline{B}}\right) \cdot \left(\overline{B} + \overline{\overline{A}}\right)$$

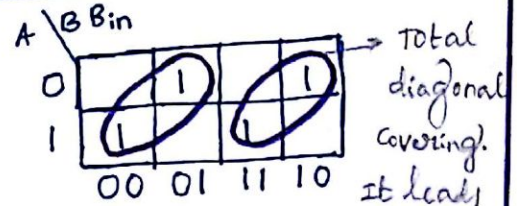
$$= \left(\overline{A} + B\right) \cdot \left(\overline{B} + A\right)$$

$$= \overline{A}\,\overline{B} + \overline{A}A + B\overline{B} + AB \quad (\overline{A}A = 0,\ B\overline{B} = 0)$$

$$= AB + \overline{A}\,\overline{B}$$

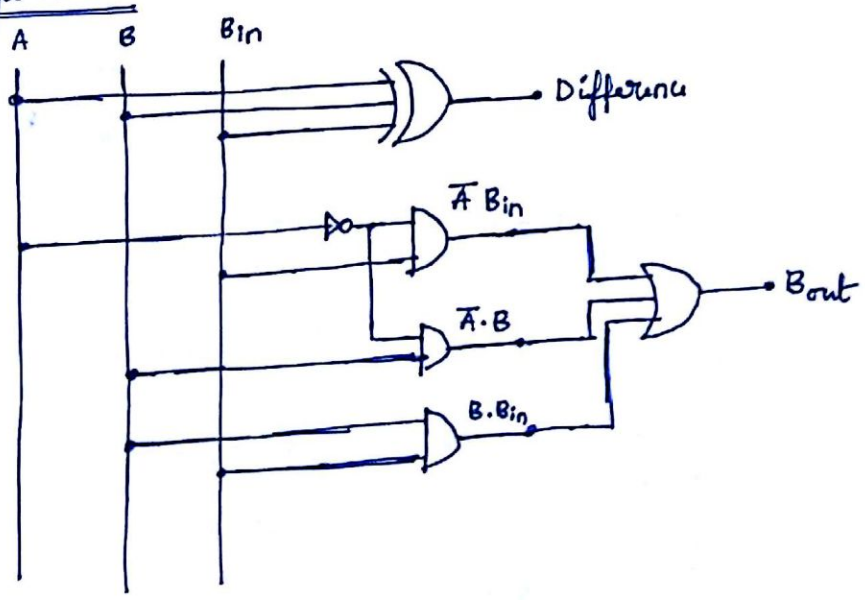
$$\boxed{\overline{A \oplus B} = AB + \overline{A}\,\overline{B}}$$

Alternative Method / shortcut



$$= A \oplus B \oplus B_{in}$$

* **Logic circuit**



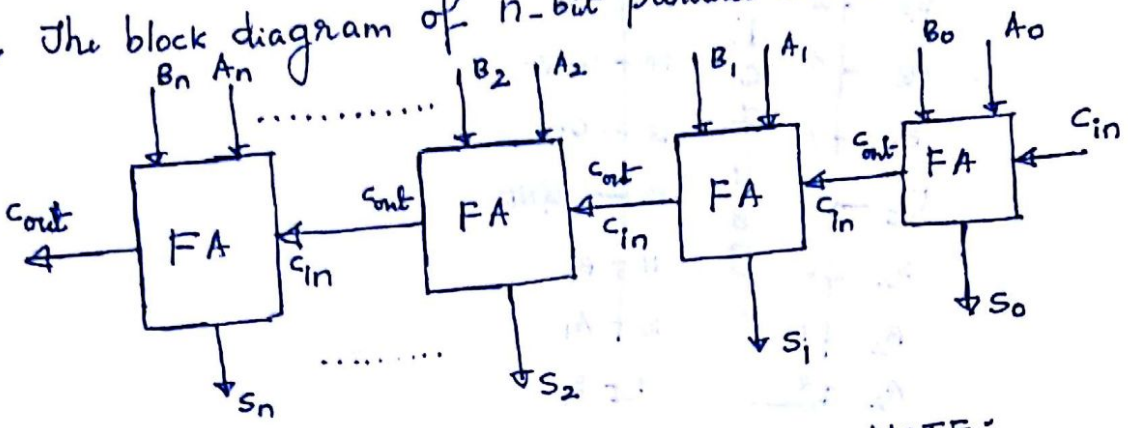
Scanned by CamScanner

Dr. K BR

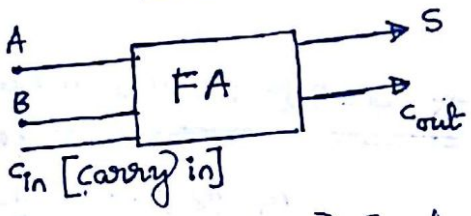
* parallel Adder [cascading full adders] :-

* N-Bit parallel Adder :

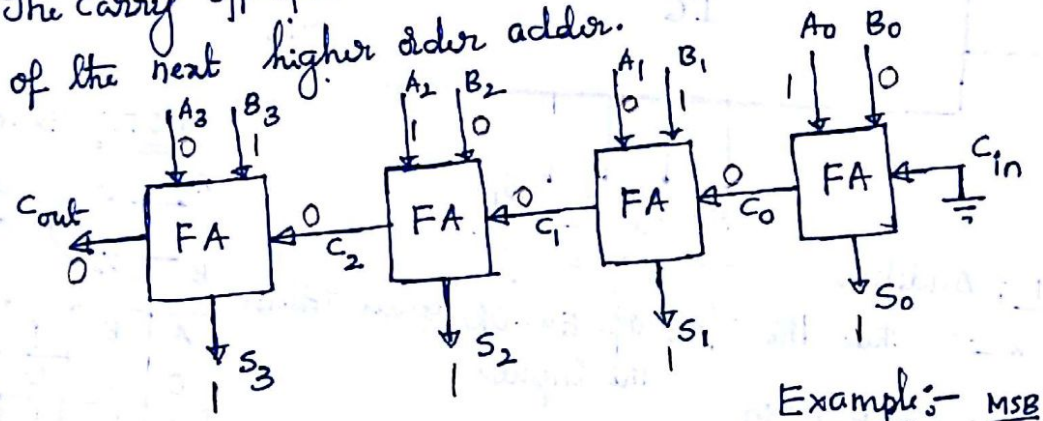
→ The block diagram of n-bit parallel adder is shown below



NOTE:



↳ The carry o/p of each adder is connected to the carry input of the next higher order adder.



Example:- MSB LSB

A = 0 1 0 1 → A

B = 1 0 1 0

↓

$B_3 B_2 B_1 B_0$

A = 0 1 0 1 (5)

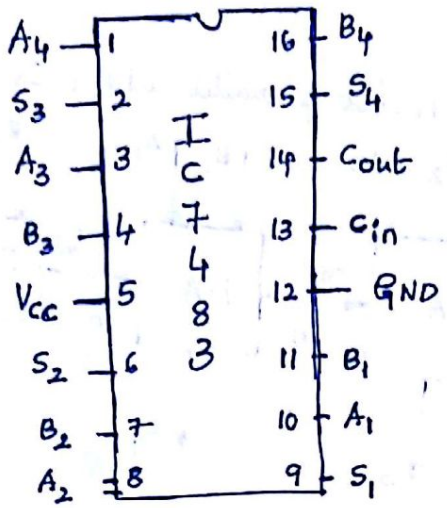
B = 1 0 1 0 (10)

1 1 1 1 (15)

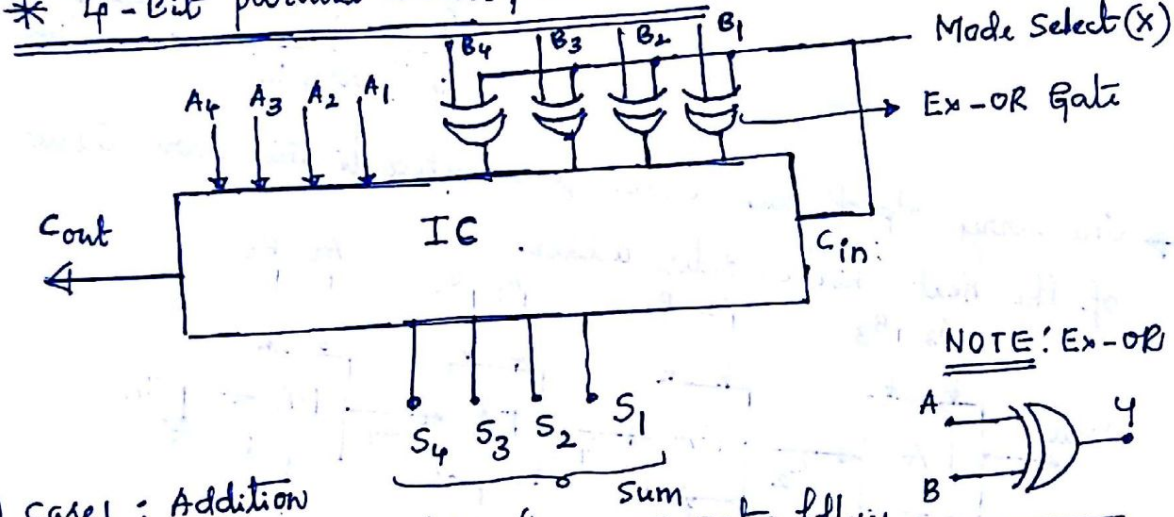
NOTE: For least significant position, carry input of Full adder is made 0.

Scanned by CamScanner

* IC 7483 :



* 4-Bit parallel Adder / Subtractor :



NOTE: Ex-OR

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

1) Case1 : Addition

if $x = 0$ then the O/P's of Ex-OR gates follows the inputs

$Sum = A + B + C_{in}$

$= A + B + 0 = A + B$

$\therefore$ $A_4\ A_3\ A_2\ A_1$

$+\ B_4\ B_3\ B_2\ B_1$

$Sum = S_4\ S_3\ S_2\ S_1$

Ex:

$1\ 1\ 1$

$1\ 1\ 0\ 1$

$1\ 0\ 1\ 1$

$\boxed{1}\ 1\ 0\ 0\ 0$

C_4

$1\ 1\ 0\ 0\ 0$

$C_{out}\ S_4\ S_3\ S_2\ S_1$

Observation :

if $B = 0$ then $Y = A$

if $A = 0$ then $Y = B$

if $B = 1$ then $Y = \overline{A}$

if $A = 1$ then $Y = \overline{B}$

Scanned by CamScanner

Dr. K B R

2) Case 2: Subtraction

If X = 1 then the o/p's of Ex-OR gates is the complement of i/p's.

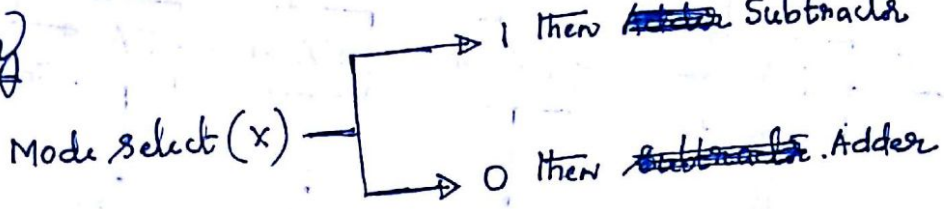
$\therefore$ Sum = A + (one's complement of B) + C_{in}

= A + 1's complement of B + 1

= A + 2's complement of B

o/p = A − B

Summary



$\therefore$ NOTE : 2's complement method is used for subtraction

Example for subtraction :

(A) 1 1 0 1

(B) 1 0 1 1 —1's complement→ 0 1 0 0

+1 0 0 0 1

0 1 0 1

+(A) 1 1 0 1

1 0 0 1 0

↑ Complement the carry

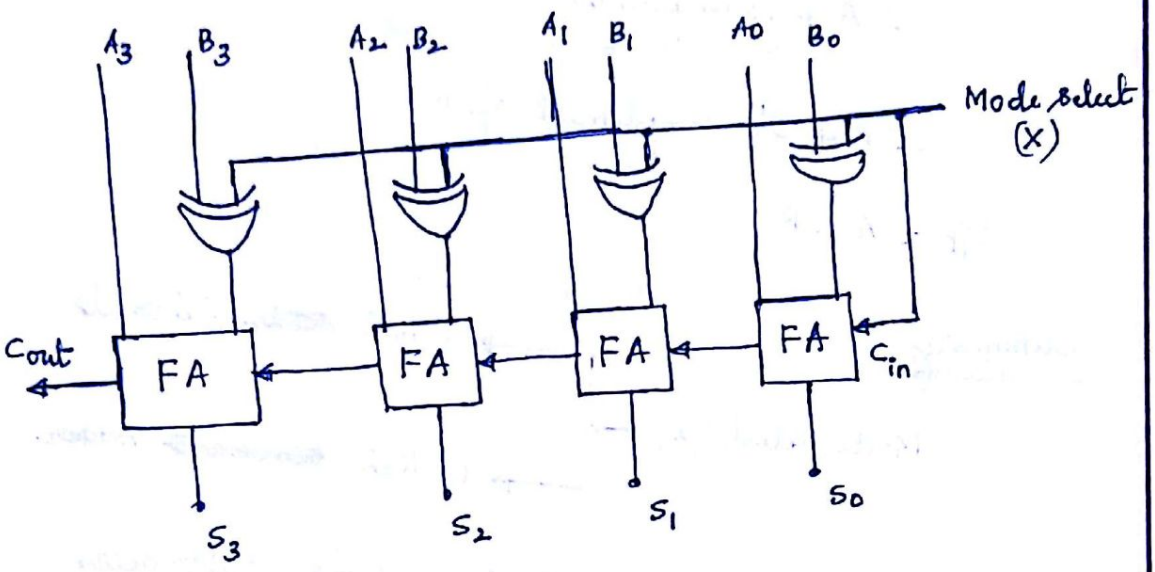
Scanned by CamScanner

* Parallel Adder / Subtractor

Example $A_3\ A_2\ A_1\ A_0$

A = 1 1 0 1

B = 1 0 1 1 → B_0, B_1, B_2, B_3



Answer Verification:

1) Adder

A = 1 1 0 1

+ B = 1 0 1 1

1 0 0 0 0

S_4 S_3 S_2 S_1 S_0

Cout

2) Subtractor

A = 1 1 0 1

B = 1 0 1 1

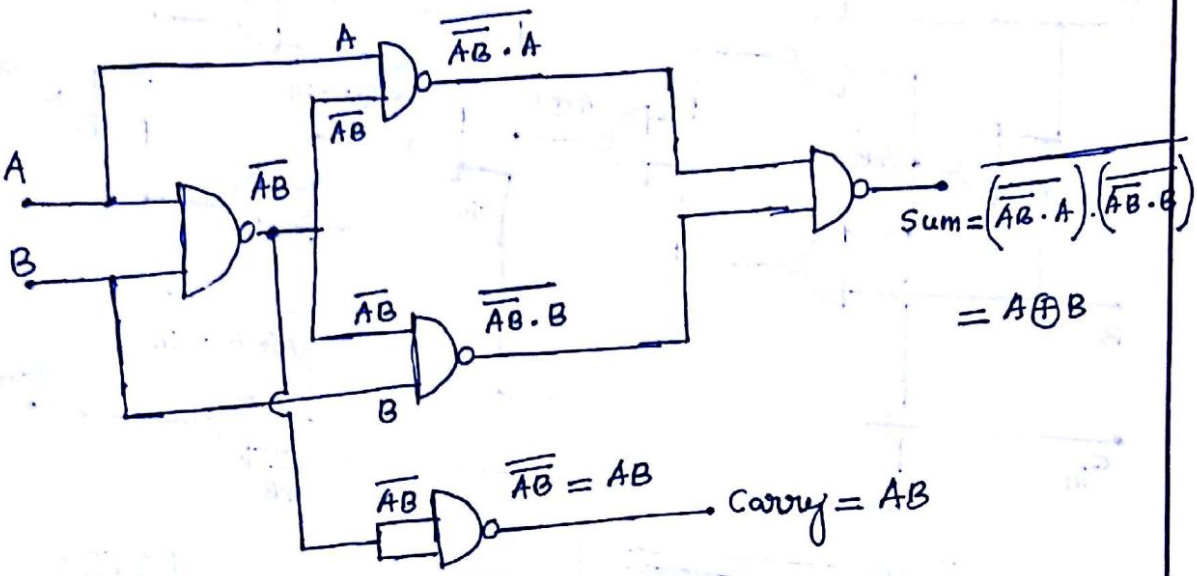
1 0 0 1 0

Cout S_3 S_2 S_1 S_0 After 2's complement

Scanned by CamScanner

Dr. KBR

* Half Adder using NAND Gates:



$$\text{Sum} = \overline{\left(\overline{AB}\cdot A\right)}\cdot\overline{\left(\overline{AB}\cdot B\right)}$$

$$= \left(\overline{\overline{AB}\cdot A}\right) + \left(\overline{\overline{AB}\cdot B}\right)$$

$$= \overline{AB}\cdot A + \overline{AB}\cdot B$$

$$= \overline{AB}\left(A+B\right)$$

$$= \left(\overline{A}+\overline{B}\right)\left(A+B\right)$$

$$= A\overline{A} + B\overline{A} + A\overline{B} + B\overline{B}$$

($A\overline{A}$ and $B\overline{B}$ → 0)

$$= A\overline{B} + B\overline{A}$$

$$= A \oplus B$$

NOTE:

$\overline{XY} = \overline{X} + \overline{Y}$

$\overline{X+Y} = \overline{X}\cdot\overline{Y}$

$X\cdot\overline{X} = 0$

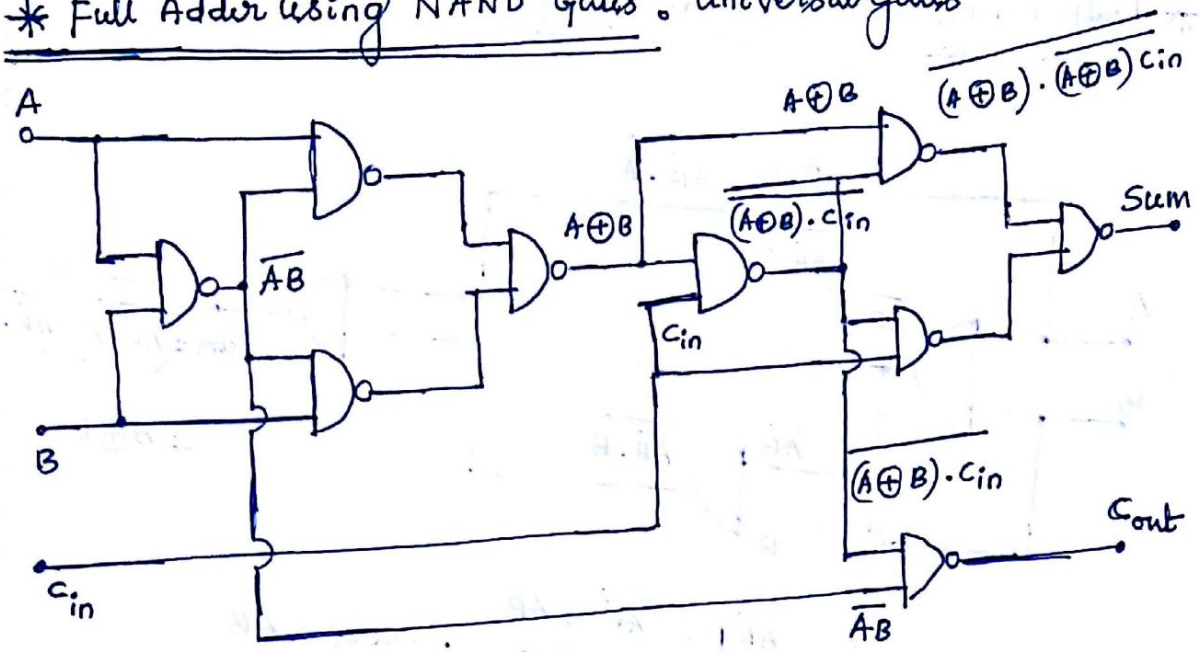
if $X = 0$, $\overline{X} = 1$

$0 \cdot 1 = 0$

NOTE: 5 (five) NAND gates are required to implement Half adder circuit

Scanned by CamScanner

* Full Adder using NAND Gates : universal gates



$$C_{out} = \left[\overline{(\overline{AB}) \cdot \overline{(A \oplus B) \cdot C_{in}}} \right]$$

$$= \overline{\overline{AB}} + \overline{\overline{(A \oplus B) \cdot C_{in}}}$$

$$C_{out} = AB + C_{in}(A \oplus B)$$

NOTE:

$$\overline{xy} = \bar{x} + \bar{y}$$

Scanned by CamScanner

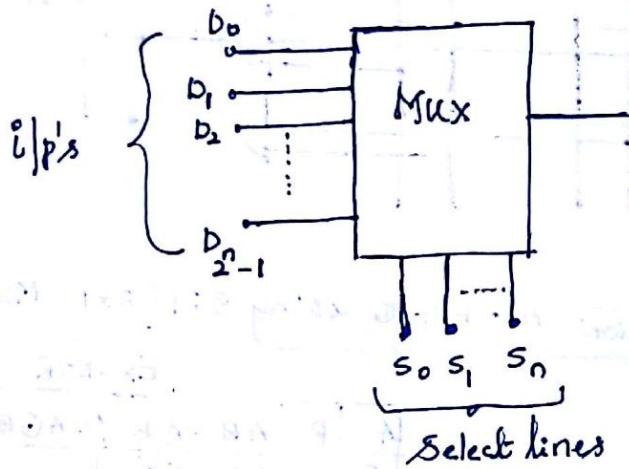
Dr. K B Ramesh

Multiplexer (Mux)

→ Multiplexers is a combinational logic circuit used to select only one input among several inputs based on selection lines

→ This circuit act as digital switch / Data selector.

→ This circuit has 2^n inputs; n select lines and only one o/p.



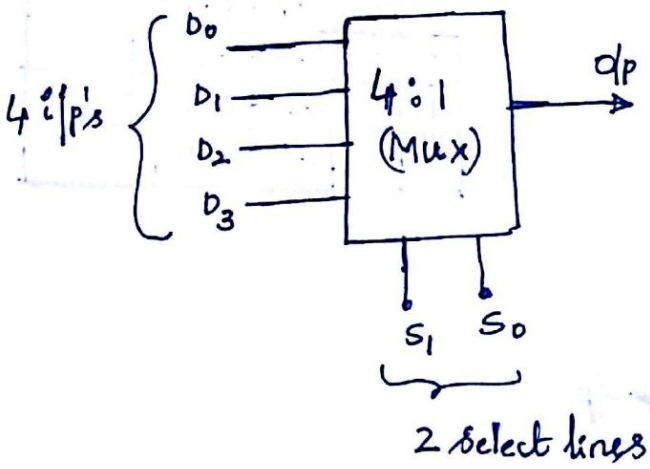
Example:

For 8:1 Mux; $2^3 = 8$ → select lines

For 4:1 Mux; $2^2 = 4$

For 16:1 Mux; $2^4 = 16$

* Design of 4:1 Mux:



Truth Table

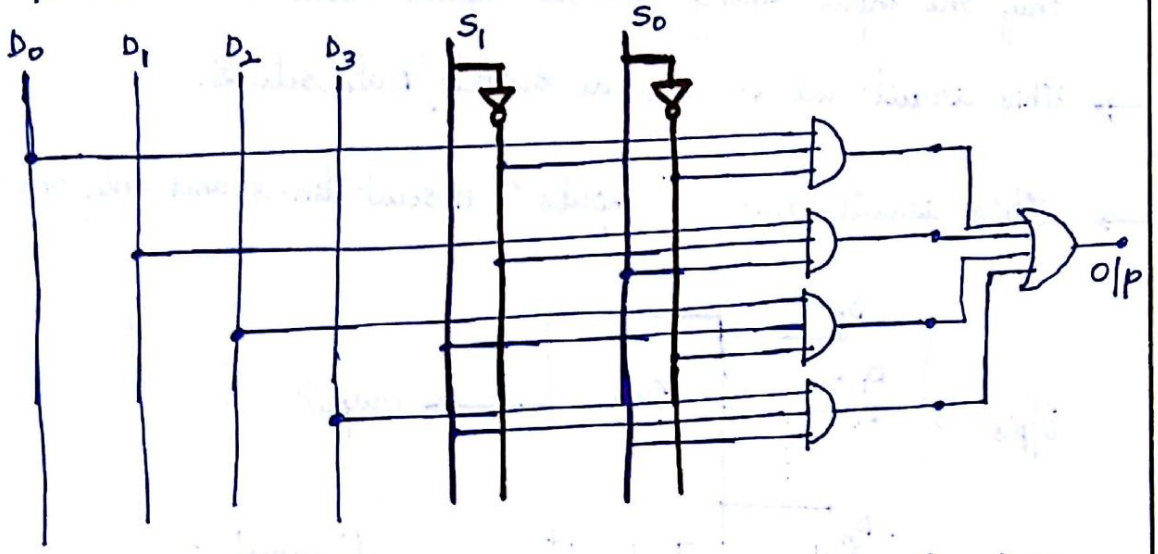
S_1	S_0	o/p
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

$$o/p = D_0 \bar{S_1} \bar{S_0} + D_1 \bar{S_1} S_0 + D_2 S_1 \bar{S_0} + D_3 S_1 S_0$$

Scanned by CamScanner

Logic circuit :

$$O/p = D_0 \bar{S_1} \bar{S_0} + D_1 \bar{S_1} S_0 + D_2 S_1 \bar{S_0} + D_3 S_1 S_0$$



e) Implement the Logic function $A \odot B \oplus C$ using 8:1 (8×1) Mux :

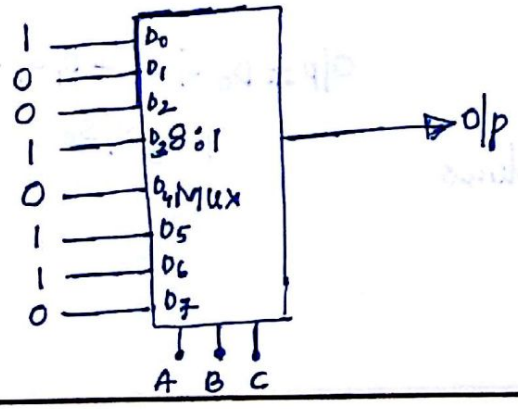
A	B	C	$f = A \odot B \oplus C$
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Ex-NOR

A	B	$AB + \bar{A}\bar{B}$	$Y = A \odot B$
0	0	$0.0 + \bar{0}.\bar{0}$	1
0	1	$0.1 + \bar{0}.\bar{1}$	0
1	0	$1.0 + T.\bar{0}$	0
1	1	$1.1 + T.\bar{T}$	1

Ex-OR

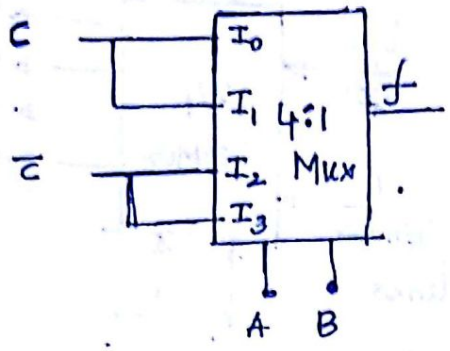
A	B	$A.\bar{B} + B.\bar{A}$	$Y = A \oplus B$
0	0	$0.1 + 0.1$	0
0	1	$0.0 + 1.1$	1
1	0	$1.1 + 0.0$	1
1	1	$1.0 + 1.0$	0



Scanned by CamScanner

Dr. K. B. Ramesh

p) The logic Realized by the circuit shown below.



a) $f = A \odot C$

b) $f = A \oplus C$ ✓

c) $f = B \odot C$

d) $f = B \oplus C$

Solution:

A	B	f
0	0	C
0	1	C
1	0	$\overline{C}$
1	1	$\overline{C}$

$$f = C.\overline{A}\,\overline{B} + C\overline{A}.B + \overline{C}.A.\overline{B} + \overline{C}.A.B$$

$$= \overline{A}.C(B+\overline{B}) + \overline{A}\,\overline{C}(\overline{B}+B)$$

$$= \overline{A}C + A.\overline{C}(B+\overline{B})$$

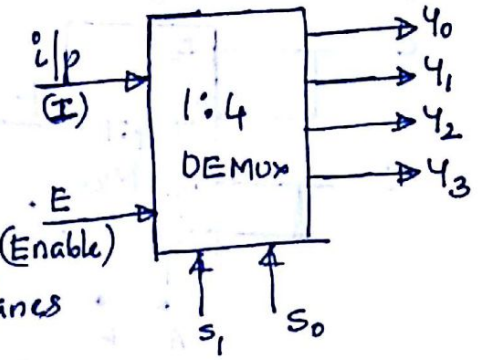
$$= \overline{A}C + A\overline{C}$$

$$\boxed{f = A \oplus C} = A \oplus C$$

Scanned by CamScanner

Demultiplexer : (1:4) Demultiplexer

- is a combinational circuit
- one input and many outputs
- Reverse operation of Mux
- Also called as Data distributor



m → No. of select lines

$n = 2$

n → No. of output lines

$$4 = 2^2$$

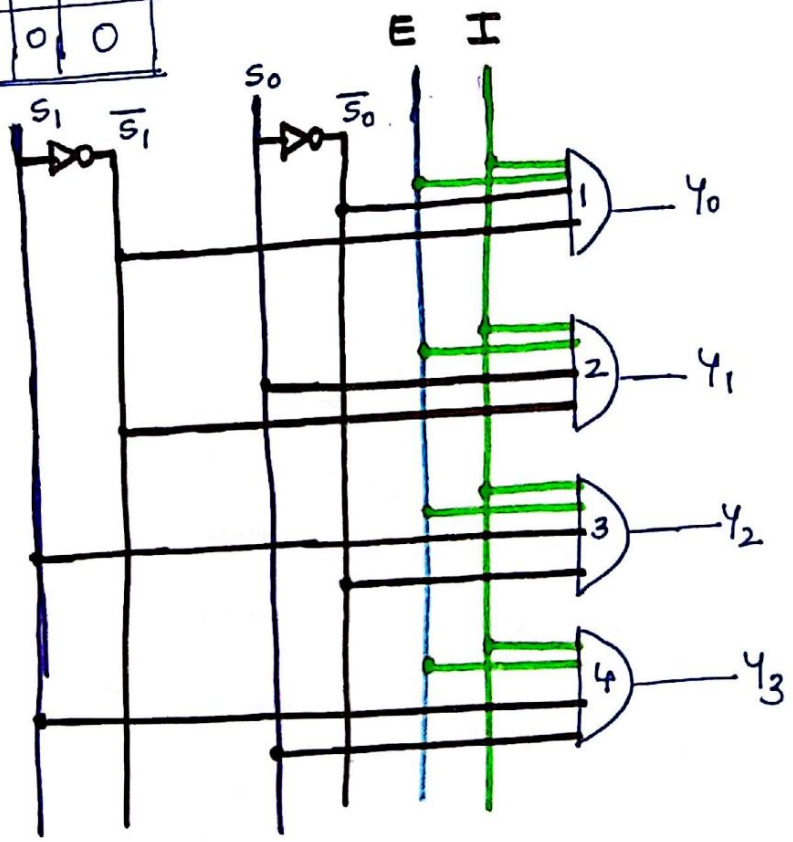
E	S_1	S_0	Y_3	Y_2	Y_1	Y_0
0	X	X	0	0	0	0
1	0	0	0	0	0	I
1	0	1	0	0	I	0
1	1	0	0	I	0	0
1	1	1	I	0	0	0

$Y_0 = E\,I\,\bar{S_1}\,\bar{S_0}$

$Y_1 = E\,I\,\bar{S_1}\,S_0$

$Y_2 = E\,I\,S_1\,\bar{S_0}$

$Y_3 = E\,I\,S_1\,S_0$

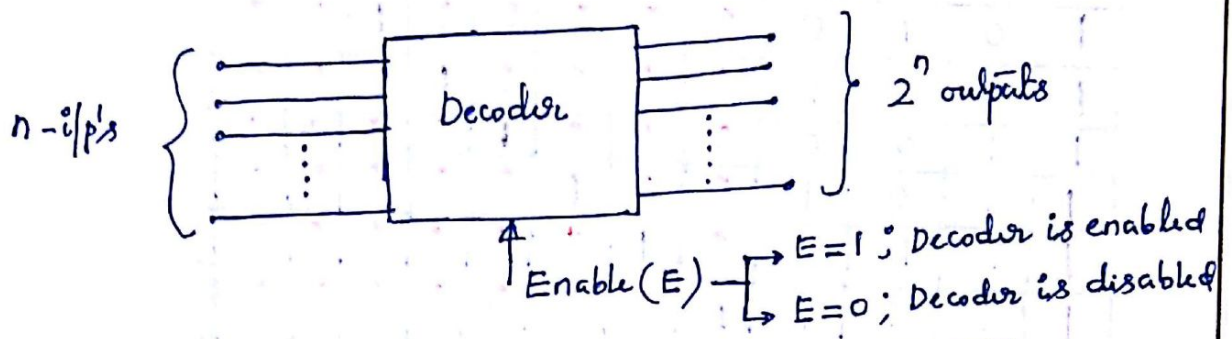


Scanned by CamScanner

Dr. K. B. Ramesh

* Decoders:

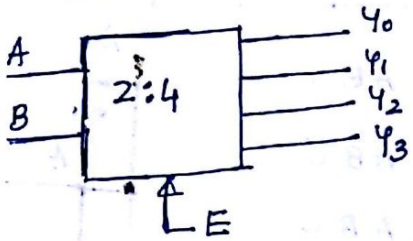
→ Is a multi-input and multi-output logic circuit which decodes n-inputs into 2^n possible outputs



Example:

n-i/p's	o/p's: 2^n
2	$2^2=4$
3	$2^3=8$
4	$2^4=16$

2 to 4 Decoder



	n i/p's		2^n - o/p's			
			2^3	2^2	2^1	2^0
E	A	B	Y_3	Y_2	Y_1	Y_0
0	X	X	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

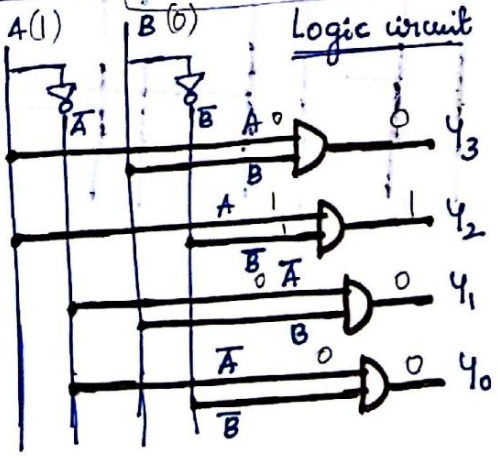
Zero → 0 → placed at 2^0 place
1 → placed at 2^1 place
1 → placed at 2^2 place
1 → placed at 2^3 place

$Y_3 = AB$

$Y_2 = A\bar{B}$

$Y_1 = \bar{A}B$

$Y_0 = \bar{A}\bar{B}$



NOTE: We can apply K-map only when there is more than one 1's in each column. Here, we have only one in each column. ∴ K-map is not required

Scanned by CamScanner

					2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
	E	A	B	C	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
1→	1	0	0	0	0	0	0	0	0	0	0	1
1→	1	0	0	1	0	0	0	0	0	0	1	0
	1	0	1	0	0	0	0	0	0	1	0	0
	1	0	1	1	0	0	0	0	1	0	0	0
	1	1	0	0	0	0	0	1	0	0	0	0
	1	1	0	1	0	0	1	0	0	0	0	0
	1	1	1	0	0	1	0	0	0	0	0	0
	1	1	1	1	1	0	0	0	0	0	0	0
0→	0	X	X	X	0	0	0	0	0	0	0	0

$Y_7 = ABC$

$Y_6 = AB\bar{C}$

$Y_5 = A\bar{B}C$

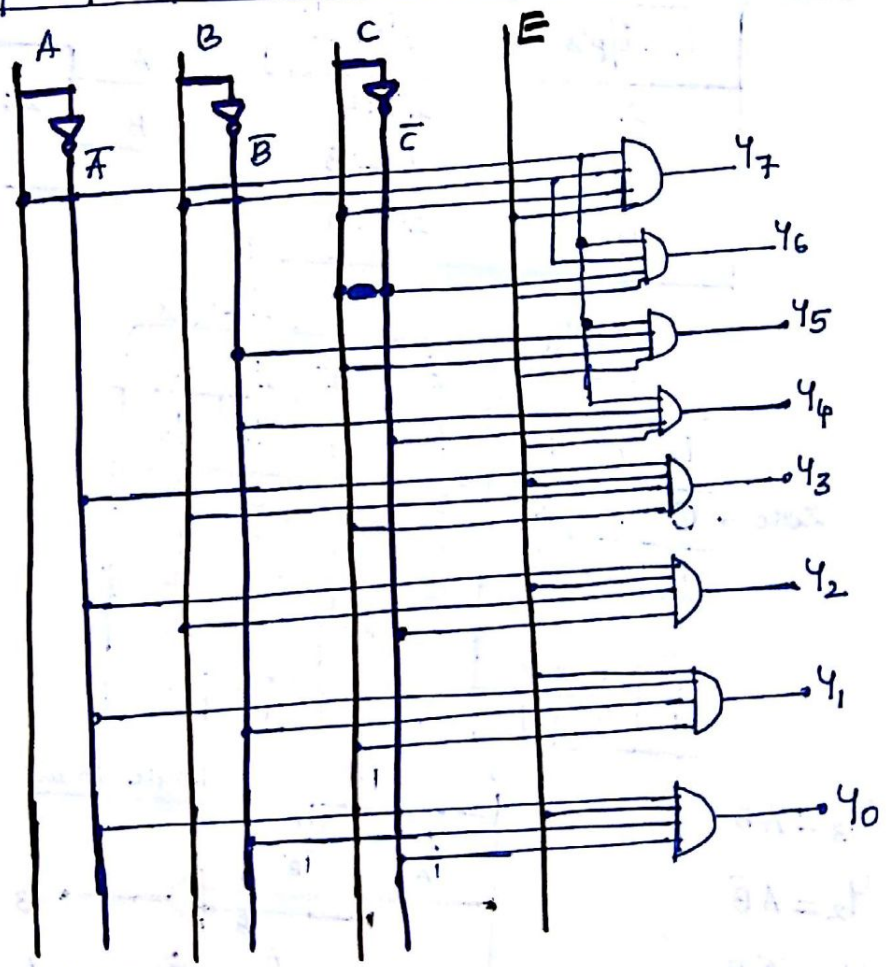
$Y_4 = A\bar{B}\bar{C}$

$Y_3 = \bar{A}BC$

$Y_2 = \bar{A}B\bar{C}$

$Y_1 = \bar{A}\bar{B}C$

$Y_0 = \bar{A}\bar{B}\bar{C}$

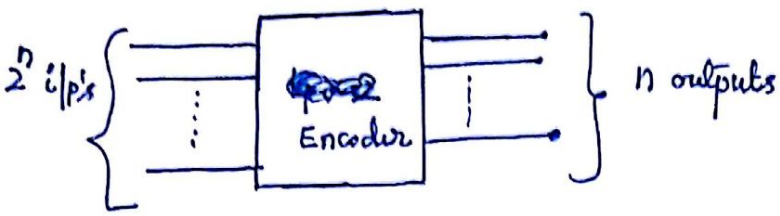


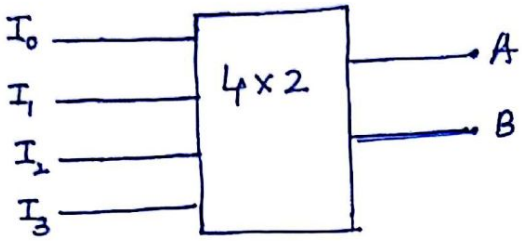
Scanned by CamScanner

Dr. K. B. Ramesh

* Design of Encoder :

4x2 Encoder

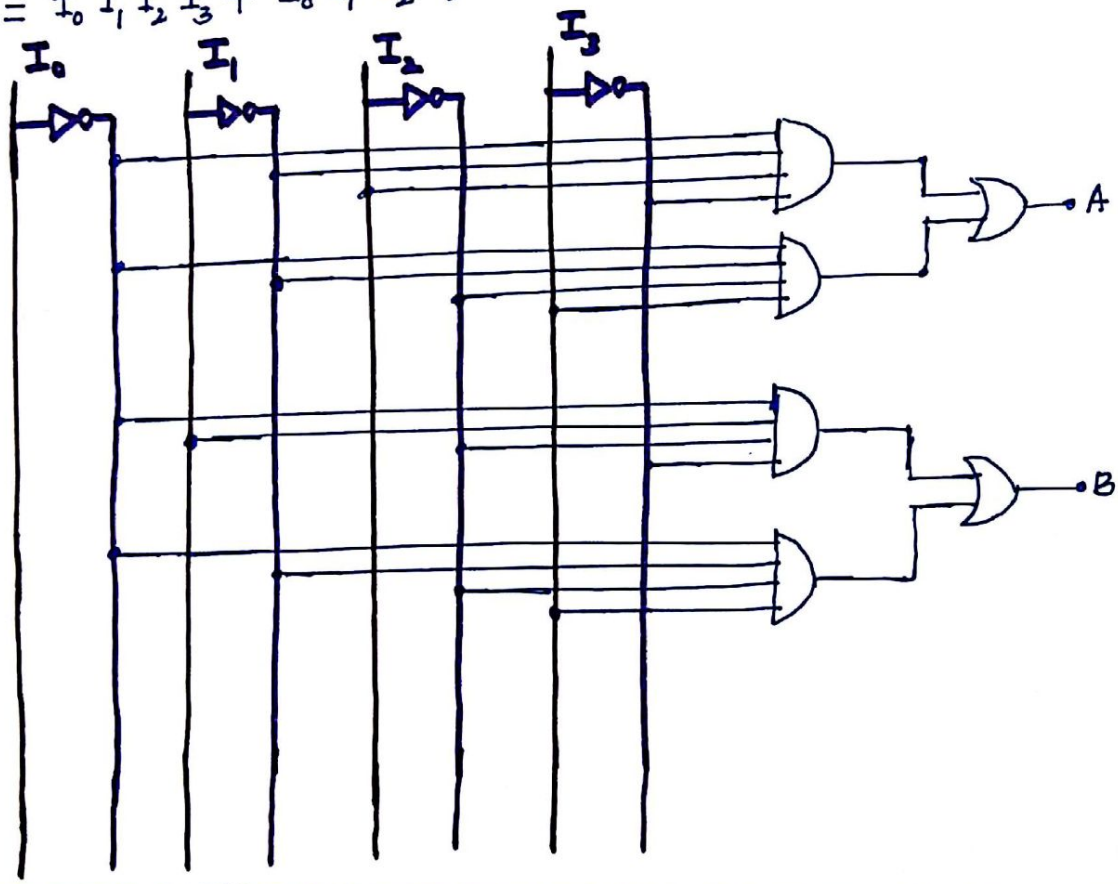




I/P's				O/P	
I_0	I_1	I_2	I_3	A	B
1	0	0	0	0	0
0	1	0	0	0	1
0	0	1	0	1	0
0	0	0	1	1	1

$$A = \bar{I_0}\,\bar{I_1}\,I_2\,\bar{I_3} + \bar{I_0}\,\bar{I_1}\,\bar{I_2}\,I_3$$

$$B = \bar{I_0}\,I_1\,\bar{I_2}\,\bar{I_3} + \bar{I_0}\,\bar{I_1}\,\bar{I_2}\,I_3$$



Scanned by CamScanner